

Introduction to HCI

Glossary

Core HCI Concepts

Human-Computer Interaction (HCI)

The study and design of how people interact with technology, focusing on making systems usable, efficient, and enjoyable.

Interactive System

Any technology that allows real-time interaction with users (e.g., smartphones, ATMs, voice assistants).

User Interface (UI)

The visual and functional parts of a system that users interact with (buttons, screens, menus, sounds).

User Experience (UX)

The complete experience of using a product, including emotions, ease of use, and satisfaction.

Design Principles

Usability

A measure of how effectively, efficiently, and satisfactorily users can achieve goals with a product.

Components: Learnability, Efficiency, Memorability, Error Prevention, Satisfaction.

Utility

Whether a product's functions meet users' needs (e.g., a translation app that accurately translates).

Likeability

How pleasurable or engaging a product feels, beyond pure functionality.

Human-Centered Design

Design approach prioritizing user needs through research, prototyping, and testing.

Interaction Design (IxD)

Designing interactive behaviors (e.g., how a button responds when clicked).

5 Dimensions of IxD

1. **Words** (Text clarity)
 2. **Visuals** (Layouts, icons)
 3. **Physical Objects** (Keyboards, touchscreens)
 4. **Time** (Animations, loading bars)
 5. **Behavior** (System feedback, user flow).
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Theories & Models

Task-Artifact Cycle

How human needs drive technology creation, which then reshapes needs (e.g., social media changing communication habits).

Maslow's Hierarchy of Needs

A pyramid of human motivations:

1. **Physiological** (food, shelter)
2. **Safety** (security, stability)
3. **Love/Belonging** (social connections)
4. **Esteem** (achievement, recognition)
5. **Self-Actualization** (creativity, purpose).
Example: Social apps address "Love/Belonging."

Experience Economy

Business strategy where value comes from memorable experiences (e.g., Apple's in-store workshops).

Technical Terms

GUI (Graphical User Interface)

UI with visual elements like windows and icons (e.g., computer desktop).

Voice User Interface (VUI)

Interaction via voice (e.g., Alexa, Siri).

Modality

Interaction mode (touch, voice, gestures).

ISO 9241-210

International standard for human-centered design principles.

Process & Evaluation

Prototyping

Creating early versions of a product to test ideas.

Context of Use

The environment, users, and goals affecting how a product is used (e.g., a medical app in a noisy hospital).

Heuristic Evaluation

Checking a design against usability best practices.

A/B Testing

Comparing two design versions to see which performs better.

Examples & Analogies

"Leaky Pipe" Analogy

Bad usability wastes user effort like a leaky pipe wastes water ([Reference](#)).

Sketching Interfaces

Drawing rough UI designs to explore ideas (e.g., time-zone converter sketches).

Key Figures & References

Ben Shneiderman

HCI pioneer; emphasized "direct manipulation" interfaces.

Book: Leonardo's Laptop (on human needs in computing).

Jakob Nielsen

Usability expert; defined usability components.

Gillian Crampton Smith

Defined the 4 original dimensions of interaction design.

Common Pitfalls

Invisible Innovation

Good HCI solutions seem obvious after they exist (e.g., smartphone swipe gestures).

Feature Bloat

Adding unnecessary functions that harm usability.